

Bit Manipulation I



Good
Evening ☺

"There has never been a
meaningful life built on
easy street."

~ John Paul Warren



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Content for Today :-

→ Basics of bit Manipulation

- Number system & conversions
- Addition of two binary numbers
- Bitwise operators
- Basic Properties of bitwise operators

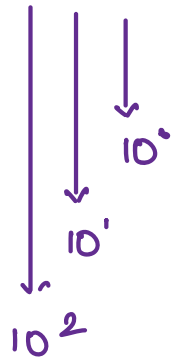
Number system → It is the way of writing to express the numbers.

01. **Decimal No. system**

Base → 10

Digits → {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

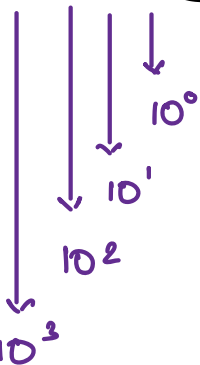
$$[743]_{10} \rightarrow 743$$



$$= 700 + 40 + 3$$

$$= \underline{\underline{743}} \quad \{ \text{digits} * \text{base value} \}$$

$$[2463]_{10} = 2 \times 10^3 + 4 \times 10^2 + 6 \times 10^1 + 3 \times 10^0$$



$$= 2000 + 400 + 60 + 3$$

$$= \underline{\underline{2463}}$$

Counting 0 1 2 3 4 5 6 7 8 9

10 11 12 13 14

. . . 99

100 101 102 . .

02. Binary no. Systems

Base - 2

Bits $\rightarrow \{0, 1\}$

$$\begin{array}{c} [0110]_2 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \end{array} = 0 * 2^3 + 1 * 2^2 + 1 * 2^1 + 0 * 2^0$$
$$= 4 + 2$$
$$= 6 \quad \{ \text{bits} * \text{base value} \}$$

Counting $\rightarrow$

0	1		
10	11		
100	101	110	111
1000			

Binary to Decimal

$$[1000]_2 \longrightarrow [8]_{10}$$

Diagram showing the conversion of binary 1000 to decimal 8. The binary digits are 1, 0, 0, 0. The weights are $2^3, 2^2, 2^1, 2^0$ respectively. The value is $1 \times 2^3 = 8$.

02. $[10100]_2 \longrightarrow 1 \times 2^4 + 1 \times 2^2$
 $= 16 + 4 = 20$

03. $[1011010]_2 = 64 + 16 + 8 + 2$
 $= [90]_{10}$

Idea $\rightarrow$ Extract the digits one by one & multiply them with their base values.

$\rightarrow$. Add all the extracted digits

```
int btod (int n) {
```

```
    ans = 0, mul = 1
```

```
    while (n > 0)
```

```
        int r = n % 10
```

```
        n = n / 10
```

```
        ans = ans + r * mul
```

```
        mul = mul * 2;
```

```
    }
```

```
    return ans;
```

```
}
```

TC = $O(\text{no. of digits})$

SC = $O(1)$

TC = $O(\log_{10} n)$

SC = $O(1)$

$$n \rightarrow \frac{n}{2} \rightarrow \frac{n}{4} \rightarrow \frac{n}{8} \dots 1 = O(\log_2 n)$$

$$n \rightarrow \frac{n}{3} \rightarrow \frac{n}{9} \rightarrow \frac{n}{27} \dots 1 = O(\log_3 n)$$

$$n \rightarrow \frac{n}{10} \rightarrow \frac{n}{100} \dots 1 = O(\log_{10} n)$$

Decimal to Binary

$$[20]_{10} \longrightarrow [10100]_2$$

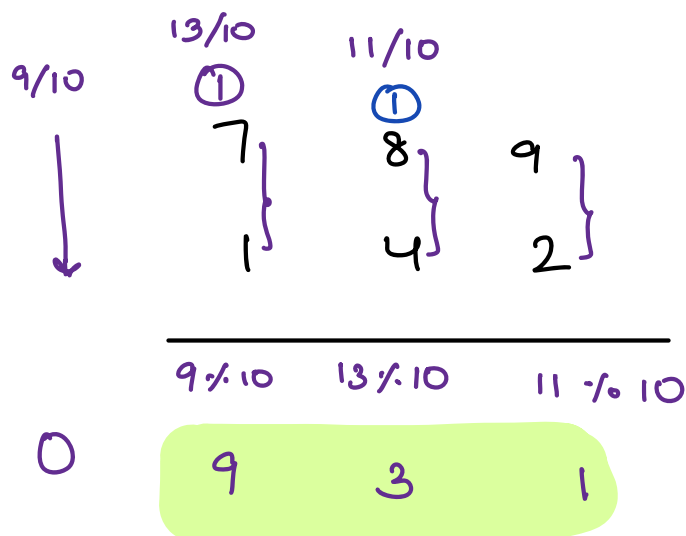
2	20	
2	10	0
2	5	0
2	2	1
2	1	0
	0	1

$$[45]_{10} \longrightarrow [101101]_2$$

2	45	
2	22	→ 1
2	11	→ 0
2	5	→ 1
2	2	→ 1
2	1	→ 0
	0	→ 1

Code → {TODO}

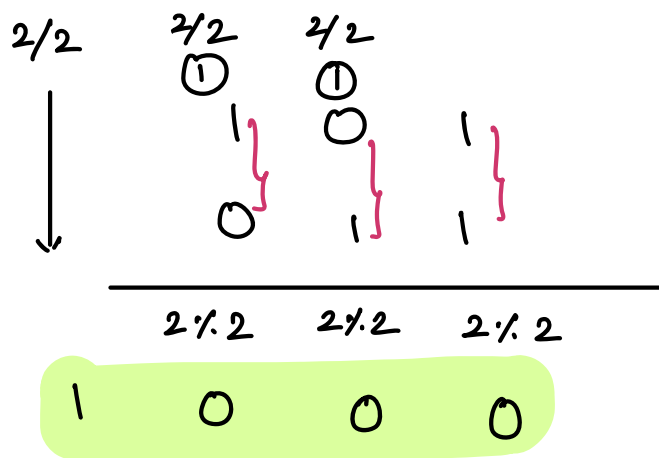
Addition of Decimal numbers



$$d = \text{sum} \% 10$$

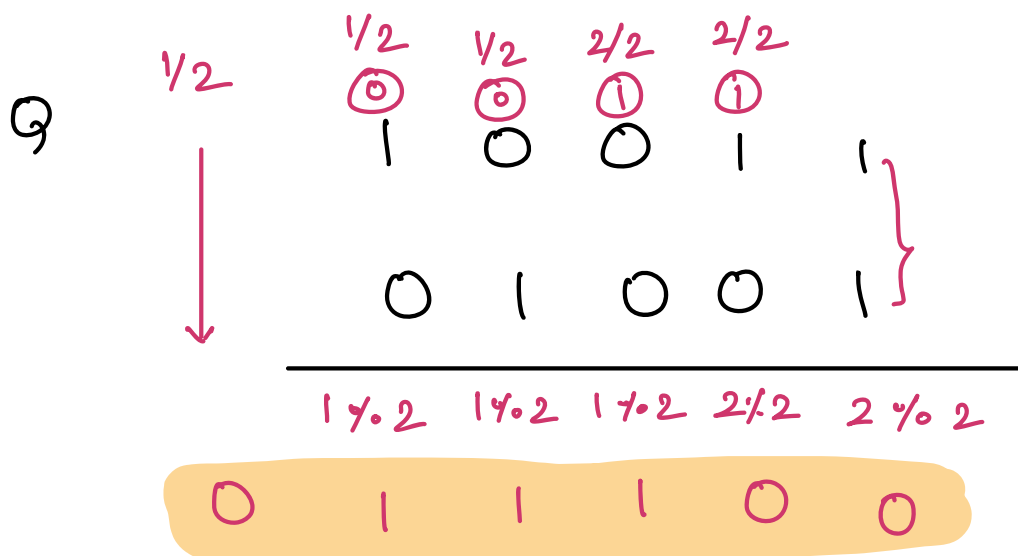
$$c = \text{sum} / 10$$

Addition of binary no.



$$d = \text{sum} \% 2$$

$$c = \text{sum} / 2$$



$$\text{sum} / 2$$

$$\text{sum} \% 2$$

n_1	n_2	$n_1 \% 10$ d_1	$n_2 \% 10$ d_2	$sum/10$ carry	$d_1 + d_2 + c$ sum	$sum \% 10$ rem	ans
789	142	9	2	0	11	1	$1 * 10^0$
78	14	8	4	1	13	3	$3 * 10^1$
7	1	7	1	1	9	9	$9 * 10^2$
0	0			0			931

Code $\rightarrow$ {TODO}

Bit Manipulation

int $\rightarrow$ 4 bytes $\rightarrow$ 32 bits

int $x = 45 \rightarrow 101101$

$x =$

<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	...	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>
21	30	29	28		6	5	4	3	2	1	0

10:16 $\rightarrow$ 10:23 pm

Bitwise operators $\rightarrow$ $\&$, $|$, $\wedge$, $\sim$, $\ll$, $\gg$
 AND OR XOR NOT Left Right

A	B	$A \& B$	$A B$	$A \wedge B$	$\sim A$	$\sim B$
0	1	0	1	1	1	0
0	0	0	0	0	1	1
1	1	1	1	0	0	0
1	0	0	1	1	0	1

0 $\rightarrow$ unset bit

1 $\rightarrow$ set bit

$A \& B \rightarrow$ Both the bits has to be set (1)

$A | B \rightarrow$ One bit has to be set (1)

$A \wedge B \rightarrow$ Same same puppy shame

A = 20

B = 45

A = 0 1 0 1 0 0

B = 1 0 1 1 0 1

print (A & B) = 0 0 0 1 0 0 $\rightarrow$ 4

print (A | B) = 1 1 1 1 0 1 $\rightarrow$ 61

$$\text{print}(A \wedge B) = 111001 \rightarrow 57$$

* Properties

01. $A \& 1 =$

$$A=10 \rightarrow 1010$$

$$A=11 \rightarrow 1011$$

$$A = 1010$$

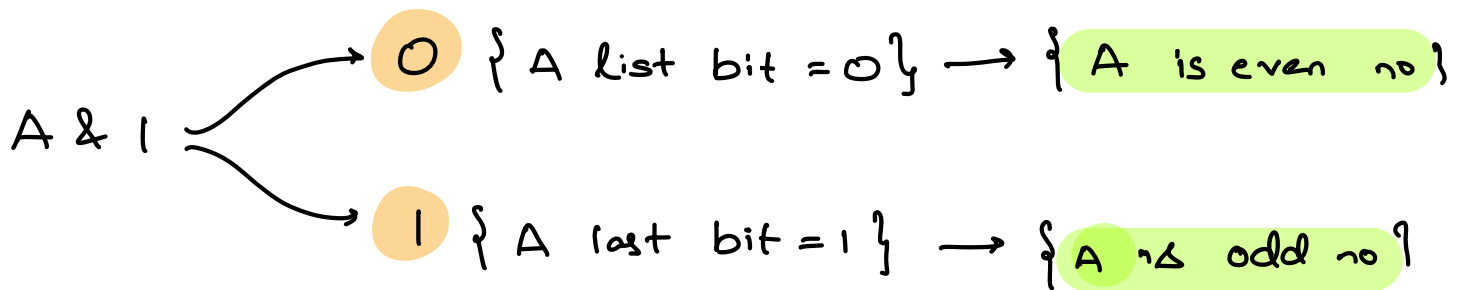
$$\& 1 = 0001$$

$$0000$$

$$A = 1011$$

$$\& 1 = 0001$$

$$0001$$



$$x = \begin{array}{cccc} & 1 & 1 & 1 & 0 \\ & \swarrow & \downarrow & \downarrow & \downarrow \\ 2^3 & & 2^2 & 2^1 & 2^0 \end{array}$$

$$= \underbrace{(1 * 2^3)}_{\text{even}} + \underbrace{(1 * 2^2)}_{\text{even}} + \underbrace{(1 * 2^1)}_{\text{even}} + \underbrace{(0 * 2^0)}_{\text{odd}}$$

even + odd $\rightarrow$ odd

02 $A \& 0 = 0$

03 $A \& A = A$

04. $A | 0 = A$

05 $A | 1 \begin{cases} A \{ A \text{ is odd?} \\ !A \{ A \text{ is even?} \end{cases}$

$$\begin{array}{r} A = 1010 \\ | \quad 1 = 0001 \\ \hline 1011 \end{array}$$

$$\begin{array}{r} A = 1011 \\ | \quad 1 = 0001 \\ \hline 1011 \end{array}$$

06. $A | A = A$

07. $A \wedge 0 = A$

$$\begin{array}{r} A = 1011 \\ \wedge \quad 0 = 0000 \\ \hline 1011 \end{array}$$

$$\begin{array}{r} A = 1010 \\ \wedge \quad 0 = 0000 \\ \hline 1010 \end{array}$$

08 $A \wedge A = 0$ } XOR of two same values = 0

$$\begin{array}{r} A = 1011 \\ \wedge A = 1011 \\ \hline 0000 \\ \hline \end{array}$$

Commutative prop

01. $A \& B = B \& A$

02. $A | B = B | A$

03. $A \wedge B = B \wedge A$

04. $A \& B \& C = C \& B \& A = A \& C \& B = B \& A \& C = B \& C \& A$

$$\underbrace{A \& (B \& C)}_{x \& y}$$

$$\begin{aligned} x &= A \\ y &= B \& C \end{aligned}$$

$$= (B \& C) \& A$$

$$= (C \& B) \& A$$

$$= C \& B \& A$$

05. $A | B | C = A | C | B = B | A | C = B | C | A = \dots$

06. $A \wedge B \wedge C = A \wedge C \wedge B = B \wedge A \wedge C = B \wedge C \wedge A = C \wedge A \wedge B = C \wedge B \wedge A$

$$Q = a^b c^a d^c b = ?$$

$$a^a b^b c^c d^d = \boxed{d}$$

Q Given an array $arr[N]$. All the numbers has the frequency two except one. Find the odd element in the array

$$arr = \{6, 9, 3, 6, 9\} \rightarrow 3$$

$$arr = \{6, 9, 10, 9, 10\} \rightarrow 6$$

Idea $\rightarrow$ XOR of the complete array

```

int oddEle (int [] ar)
{
    int ans = 0
    for (int i = 0; i < n; i++)
    {
        ans = ans ^ ar[i];
    }
    return ans;
}

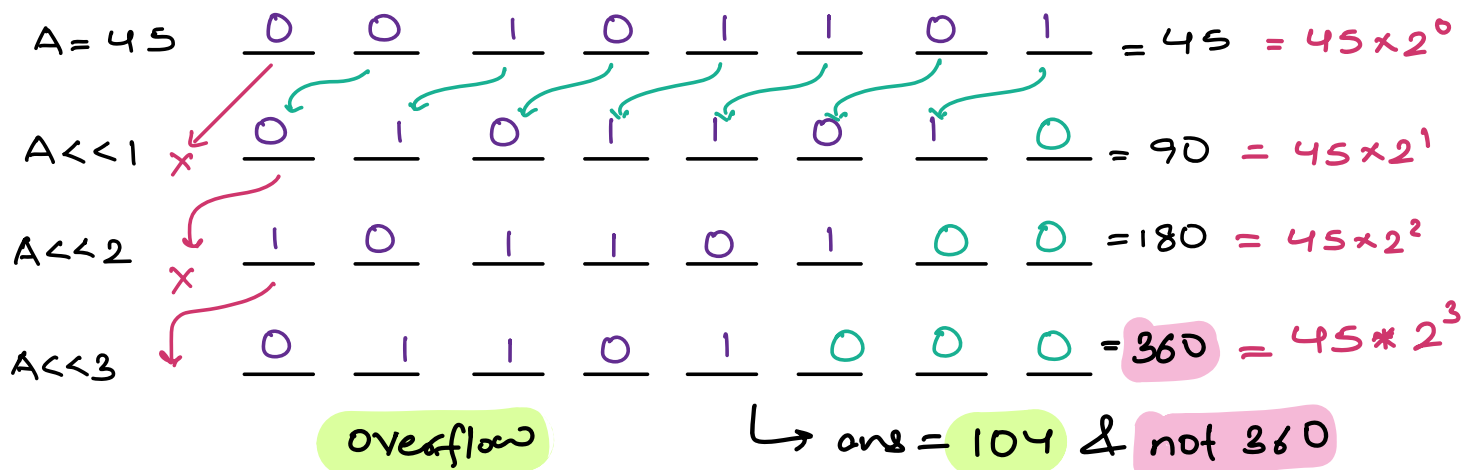
```

TC = $O(n)$
SC = $O(1)$

Left shift operator

int $\rightarrow$ 4 bytes $\rightarrow$ 32 bits

// Assumption int $\rightarrow$ 8 bits } Just for the explanation



$$a \ll n = a * 2^n$$

$$1 \ll n = 2^n$$

Max value stored by 8 bits

$$\underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} = 255$$

Right shift operator

$$A = 20 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} = 20$$

$A \gg 1 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} = 10$

A >> 2 0 0 0 0 0 1 0 1 $\times = 5$

No overflow

$$a \gg n = \frac{a}{2^n}$$

Negative no. $\rightarrow$ Will be dealt in next session

arr = {6, 9, 3, 6, 9}

6 $\rightarrow$ 0110

$9 \rightarrow 1001$

3 → 0011

$$a \sim b = \bigcirc \bigcirc \bigcirc \bigcirc$$

0 1 1 0

→ 0 1 1 0

1 0 0 1

ang $\rightarrow$ 1 1 1 1

0 0 1 1

$03 \rightarrow 1100$

0 1 1 0

$a_3 = 1010$

$$\begin{array}{r} 1001 \\ \hline 0011 \end{array} \rightarrow \boxed{3}$$

1 byte = 8 bits

int $\rightarrow$ 4 bytes = $4 * 8 = \underline{\underline{32 \text{ bits}}}$